

choptuik_ac_bc · theorems12 · github.com/wild8highlander/choptuik_ac_bc

The Clock-Pair Theorem: the exact GCD of the books on the static chain

$\tau^* = 27/4$ (baseline) · $\tau^* = 1323/(196 - \pi^2)$ (one brick) · SymPy, exact

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Abstract

This monograph records the first exact theorem of the symbolic layer of the direct solution programme for the Choptuik problem: the clock-pair theorem on the static chain of the center. It is established that the two principal clock books of the truncated tower — the UV book and the mass-defect book — are proportional identically ($UV = 2R1h \cdot M_{\text{def}}$) and therefore define one clock equation in two branches. The greatest common divisor of the pair is computed by SymPy in exact arithmetic: at the baseline point it equals $T0h^2(4T0h^2 - 27)/4$ with the single positive root $\tau^* = 27/4 = 6.75$, and on the one-brick correction $\delta_C = \pi/7$ it equals $T0h^2(\pi - 14)(\pi + 14)((196 - \pi^2)T0h^2 - 1323)/(196 - \pi^2)$ with the root $\tau^* = 1323/(196 - \pi^2) \approx 7.10792$.

Numerical tables from the JSON verdict are given (the books at x^* with defect 0.0, the station ladder, the v16 landing audit 0/36 and the honest static map 0/18), together with the corollaries — the dissolution of the naive incompatibility 9/4 versus 9/16, the one-brick shift of the root and the identity of the two independent routes — as well as the honest caveats and the complete reproduction protocol by the repository machines.

Keywords: Choptuik problem, static chain, clock pair, GCD, $\tau^* = 27/4$, $1323/(196 - \pi^2)$, SymPy, machine proof, truncated tower.

Table of Contents

1. Introduction.....	1
2. Notation and Setting.....	2
2.1. The static chain and the books.....	2
2.2. The point x^* and the critical scale.....	2
3. Statement of the Theorem.....	3
4. Sketch of the Machine Proof.....	4
4.1. Exact arithmetic.....	4
4.2. Step 1: proportionality of the books.....	4
4.3. Step 2: GCD factorization.....	5
4.4. Step 3: the books at x^* and degeneracy outside x^*	5
5. Numerical Results.....	6
5.1. The books at the critical point.....	6
5.2. The v16 revision (erratum) and the honest static map.....	6
6. Corollaries.....	7
7. Honest Caveats.....	8
8. Reproducibility.....	9
References.....	10

Note: the Table of Contents is generated via field codes. After editing, refresh the page numbers: right-click the TOC and select "Update Field".

1. Introduction

This monograph opens the “Twelve Theorems” cycle of the theorems12 folder of the choptuik_ac_bc repository and records, as a separate self-contained document, the first exact result of the symbolic layer of the direct solution programme for the Choptuik problem: the clock-pair theorem on the static chain of the center. The question is fundamental for the whole echo mechanics: how many independent clock candidates are produced by the two principal “books” of the truncated tower — the UV book and the mass-defect book — and does the pair possess a single common root, that is, a common scale τ^* at which both books ring simultaneously.

Historically, the result matured across two campaigns. In v17 (the machine clock_closure_t1c.py, LEDGER §28) a revision of the earlier static landings was carried out and the exact SymPy factorization of the greatest common divisor of the two clock equations on the T1c source form was obtained for the first time. In v18 (the global static census, LEDGER §29) the factorization was rediscovered independently as theorem [G1a] and supplemented by a systematic scan. Finally, in OPEN9 v19 question No. 5 confirmed the identity $\tau^*(\delta_C) = \tau^*_{T1c}$ — two independent routes to one and the same number. All three layers are consistent, and it is their totality that constitutes the content of the present theorem.

The relevance of the exact formulation is determined by the fact that the naive adiabatic treatment gave an incompatibility of the two branches: $\tau_3 = 9/4$ versus $1/\tau_5 = 9/16$. This incompatibility long prevented closing the clock through the truncated tower. The clock-pair theorem shows that the incompatibility is an artifact of adiabatics: after de-adiabatization of the sources both branches merge into one pair of equations with a single common root, and one brick — the only surviving correction $\delta_C = \pi/7$ — shifts the root from $27/4 = 6.75$ to $1323/(196-\pi^2) \approx 7.10792$, closing $\ln(\tau^*)$ with a remainder of +0.062% (jointly with the holonomy theorem, monograph No. 7).

The monograph follows the standard of the series: an exact statement, a sketch of the machine proof, complete numerical tables from the JSON verdict, corollaries, honest caveats and a reproduction protocol. Every number appearing below is taken from the

machine files of the repository and was not fitted to literature values; each literature anchor is labelled a “comparison anchor”.

2. Notation and Setting

2.1. The static chain and the books

By the static chain of the center (denoted C) we understand the one-parameter family of stationary configurations of the truncated tower of central amplitudes on which the residues of the tower equations reduce to exact chain forms. The chain forms express the senior amplitudes through the base height $T0h$ and the linear scale $R1h$; for the baseline point ($\kappa = 2$) they read $R3h = 2R1h \cdot T0h^2/9$, $W2h = 4T0h^2/3$, $P2h = T0h/3$, $R5h = 2R1h \cdot T0h^2(3T0h^2 - 5)/45$, $M5h = 2R1h \cdot T0h^2(23 - 6T0h^2)/9$, $D0h = 0$. These forms are reproduced by the machine identically and serve as the “stage” for all static theorems of the series.

By books (clock books) we mean functionals of the residues whose vanishing is interpreted as the “ringing of the clock” — the coincidence of the configuration scale with the critical one. The two principal books are the UV book, obtained from the third UV residue, and the mass-defect book $Mdef_{\xi 5}$, obtained from the fifth defect residue. The ring book $W2/T0^2$ measures the ratio of the second amplitude $W2h$ to the squared height and equals $4/3$ on the CSS ring; in the corrected (holonomy) system its value is $2\kappa/3$.

2.2. The point x^* and the critical scale

The critical point x^* of the tower is defined by the booking values $T0h = 3\sqrt{3}/2$, $R3h = 3/2$, $W2h = 9$, $P2h = \sqrt{3}/2$, $R5h = 183/40$, $M5h = -105/4$, $P4h = 153\sqrt{3}/10$; the baseline scale is $\tau^* = 27/4$. The point x^* is the unique non-degenerate root of the constraint system on the chain (established by the factorization theorem, monograph No. 3, and by the Krawczyk census, monograph No. 9). The symbol τ denotes the dimensionless log-time scale, and κ the dimensionless constant of the corrected system, $\kappa = 2$ at the baseline and $\kappa = 2 - \pi^2/98$ on the surviving brick.

Table 1.1. Principal notation of the monograph

Symbol	Meaning	Baseline value
T0h	height of the static chain	$3\sqrt{3}/2 \approx 2.59808$
R1h	linear scale (the ruler)	1 (at x^*)
τ^*	critical scale (clock root)	$27/4 = 6.75$
κ	constant of the corrected system	$2; 2 - \pi^2/98$ (brick)
UV_ξ3	UV book (third residue)	—
Mdef_ξ5	mass-defect book (fifth residue)	—
δ_C	surviving brick (spinor sector bC)	$\pi/7$

The problem is posed as follows: to prove that on the chain C the clock pair (UV_ξ3, Mdef_ξ5) is consistent, has exactly one common positive root, to write out this root in closed form for the baseline and one-brick constants, and to establish the exact status of both books at the critical point x^* . Consistency of the pair is understood in the strict sense of polynomial system theory: through the greatest common divisor of the two polynomials after substitution of the chain forms.

3. Statement of the Theorem

Theorem 1 (the clock pair on the static chain). Let C be the static chain of the center in the T1c source form, UV_ξ3 the UV book, Mdef_ξ5 the mass-defect book, and κ the constant of the corrected system. Then the following four statements hold.

- (i) On the chain the books are proportional identically: $UV_{\xi 3} = 2 \cdot R1h \cdot Mdef_{\xi 5}$, that is, the pair defines one clock equation in two branches, not two independent conditions.
- (ii) At the baseline $\kappa = 2$ the greatest common divisor of the pair on the chain equals $T0h^2(4T0h^2 - 27)/4$; both polynomials vanish simultaneously at the single positive root $\tau^* = 27/4 = 6.75$, to which $T0h = 3\sqrt{3}/2 \approx 2.5980762$ corresponds.
- (iii) On the one-brick correction $\kappa = 2 - \pi^2/98$ ($\delta_C = \pi/7$) the greatest common divisor equals $T0h^2(\pi - 14)(\pi + 14) \cdot ((196 - \pi^2)T0h^2 - 1323)/(196 - \pi^2)$, and the pair has the single common positive root $\tau^* = 1323/(196 - \pi^2) \approx 7.1079202069$.

(iv) At the critical point x^* both books hold with zero defect: the ring book $W2h/T0h^2 = 2\kappa/3$ and the UV book $9R3h/(2R1h \cdot T0h^2) = \kappa/2$ are exact (machine defect 0.0); the station ladder $R3h = 3/2$, $W2h = 9$, $R5h = 183/40$, $M5h = -105/4$ is holonomy-invariant, and $s = 3P2h/T0h = 1$. Outside x^* the static carrier of the clock degenerates: at the half-tower corners ($R1h \rightarrow 0$) the ring amplitudes vanish and the ring clock is dead.

$$\tau^* = 27/4 = 6.75 \text{ (baseline)}, \quad \tau^* = 1323/(196 - \pi^2) \approx 7.10792 \text{ (one brick)} \quad (1.1)$$

Formulas (ii)–(iii) mean that both roots are the unique common zeros of the pair: the individual positive roots of $UV_ξ3$ and $Mdef_ξ5$ coincide pairwise (2.59807621 at the baseline; 2.6660683 on the brick), which fixes the word “pair”: one clock equation, one pair of branches, one ring.

4. Sketch of the Machine Proof

4.1. Exact arithmetic

The proof is executed by the machine `clock_closure_t1c.py` (SymPy + mpmath, working precision 50 significant digits; BLAS pinned to one thread). Polynomial expressions are built symbolically: the chain forms are substituted into the residues $UV_ξ3$ and $Mdef_ξ5$, after which SymPy computes the greatest common divisor in exact arithmetic — over $Q(\sqrt{3})$ at the baseline and over $Q(\pi)$ on the one-brick. No numerical thresholds are used in deriving formulas (1.1) and the GCD factorizations; numerical values are quoted only for control and for the tables.

4.2. Step 1: proportionality of the books

The first step is identity (i). After substitution of the chain forms the machine simplifies the difference $UV_ξ3 - 2R1h \cdot Mdef_ξ5$ and obtains an identically zero result (completeness residual 0 in the machine terminology). By the same technique, in the factorization theorem (monograph No. 3), the remaining resting equations are shown to vanish identically on the chain, so that the pair $(UV_ξ3, Mdef_ξ5)$ indeed exhausts the static system on C . The ratio of the books equals $2R1h$ up to machine zero in all control points of both signs of κ .

4.3. Step 2: GCD factorization

The second step is the computation of the GCD. At the baseline the machine returns the factorization $GCD = T0h^2(4T0h^2 - 27)/4$: the factor $T0h^2$ answers the trivial zero family, the quadratic factor $4T0h^2 - 27$ is the clock. On the brick the structure is preserved, but the clock factor acquires the spinor correction; the complete factorization is quoted in formula (1.2):

$$GCD = T0h^2(\pi - 14)(\pi + 14)((196 - \pi^2)T0h^2 - 1323)/(196 - \pi^2) \quad (1.2)$$

The machine separately verifies that the GCD vanishes at the booking root ($gcd_at_tau_star_zero = true$), that the roots are positive and unique ($gcd_positive_roots = [2.59807621]$ and $[2.6660683]$ respectively; $unique_common_clock = true$), and that the individual positive roots of both books coincide. The check is performed both symbolically and in 50-digit numerical form with complete agreement.

4.4. Step 3: the books at x^* and degeneracy outside x^*

The third step fixes statement (iv). At x^* the machine computes the ring and UV books: $W2h/T0h^2 = 1.333333333... = 2\kappa/3$ at $\kappa = 2$ and $1.2661931673... = 2\kappa/3$ at $\kappa = 2 - \pi^2/98$, the UV book = 1.0 and $0.9496448755... = \kappa/2$ respectively; the defects of both books are exactly 0.0 at both points. The station ladder is holonomy-invariant: $R3h = 1.5 = 3/2$, $W2h = 9.0$, $R5h = 4.575 = 183/40$, $M5h = -26.25 = -105/4$ at both points. Finally, at the three half-tower corners ($R1h = 10^{-12}$ with $T0h$ from the ladder) the machine obtains $F_max \leq 3 \cdot 10^{-12}$ and $W2h/T0h^2 \rightarrow 10^{-12} \cdot (4/3)$: the ring amplitudes vanish together with $R1h$, and the ring clock is dead.

Essentially, step 3 is separated from steps 1–2: the clock is exact as a book at x^* , but its static carrier outside x^* annihilates. This separation is the key difference of the honest v17 formulation from the retracted v16 version and the main reason the theorem is stated in four separate clauses.

5. Numerical Results

5.1. The books at the critical point

Table 1.2. The clock books at x^* (machine clock_closure_t1c.py, layer C2)

Quantity	$\kappa = 2$ (baseline)	$\kappa = 2 - \pi^2/98$ (brick)	Defect
W2h / T0h ² (ring book)	1.3333333333	1.2661931673	0.0
$2\kappa/3$ (right-hand side)	1.3333333333	1.2661931673	0.0
9R3h / (2R1h·T0h ²) (UV book)	1.0	0.9496448755	0.0
$\kappa/2$ (right-hand side)	1.0	0.9496448755	0.0
$s = 3P2h / T0h$	1.0000000000	1.0000000000	0.0
R3h (holonomy-invariant)	$1.5 = 3/2$	$1.5 = 3/2$	0.0

The defect 0.0 in the table is the machine zero of exact arithmetic, not a small threshold: the left- and right-hand values are obtained from independent symbolic expressions and coincide bitwise in the 50-digit representation. This is precisely why clauses (ii) and (iv) are called exact rather than numerical: the coincidence $W2h/T0h^2 = 2\kappa/3$ holds in the field of rational functions of κ , not only at the two control points.

5.2. The v16 revision (erratum) and the honest static map

Before statement (iv) the machine audited the earlier landings (layer A0). It is established: the v16 landings of the “compensated book” did not pin T0h ($dx[0] \neq 0$); all 36 repeated landings proved to be trivial captures of the T0h branch (35 cases) or the half-tower (1 case) with ring amplitudes of order 10^{-22} and a chain deviation up to 10^6 which the filter missed because it tested R1h only. The v16 verdict on the “static ridge” is retracted; the v16 layers not depending on the pinning (T1a, T1d, T2, the exit) are preserved unchanged.

Table 1.3. The v16 landing audit (layer A0, 36 runs controlled)

Landing class	Count	Diagnosis
Trivial T0h capture (T0h →	35	ring amplitudes $\sim 10^{-22}$;

Landing class	Count	Diagnosis
$10^{-6})$		chain_dev up to 10^6
Half-tower ($R1h \rightarrow 0$)	1	degenerate corner, the ring clock is dead
Book-like landings (the needed class)	0	—

The honest static map (layer A1) with a hard pinning of T0h at the booking stations and a multi-start over five seeds of R1h gives 0 of 18 non-degenerate static states at both points: all landings drift to the half-tower $R1h \rightarrow 0$ with F in the range $10^{-14} \dots 10^{-27}$. The book defect at pinned stations reaches $F_{\text{book}} \approx 26.67$ (station 1) and decays along the ladder, which agrees with the sawtooth profile of the pure book defect of v18a (B5: $F = 26.79$, κ -independent). Jointly, Tables 1.3 and 1.4 mean: the clock lives statically only at the critical scale.

Table 1.4. The honest static map (layer A1): 0/18 non-degenerate states

Point	Landings	Non-degenerate	Outcome class	Range of F
$\kappa = 2$	9	0	half-tower $R1h \rightarrow 0$	$10^{-14} \dots 10^{-27}$
$\kappa = 2 - \pi^2/98$	9	0	half-tower $R1h \rightarrow 0$	$10^{-14} \dots 10^{-27}$

6. Corollaries

Corollary 1.1 (dissolution of the naive incompatibility). The adiabatic pair of scales $\tau_3 = 9/4$ and $1/\tau_5 = 9/16$ is inconsistent; the de-adiabatization of the sources dissolves the conflict: both branches are one clock pair with the root $27/4$ (baseline) or $1323/(196 - \pi^2)$ (brick). The incompatibility was an artifact of the zeroed second derivatives, not a property of the tower.

Corollary 1.2 (the one-brick shift). The ratio $\tau^*(\delta_C)/\tau^*(0) = (27/4)/(1323/(196 - \pi^2))$ shows that the single surviving brick $\delta_C = \pi/7$ shifts the root by +5.27%. In the

logarithmic measure this closes $\ln(\tau^*)$ from -2.57% (baseline) to $+0.062\%$ (brick) relative to the observed constant $\kappa_{\text{obs}} = \gamma_{\text{lit}} \cdot \ln 10 / \Delta_{\text{sp}}$ — a fact pinned by the machine `s4_berry_one_brick` and reused in the holonomy theorem (monograph No. 7).

Corollary 1.3 (the identity of two routes). The root (1.1) coincides exactly with the root of the clock pair computed by the `sympy_center_o7` route (question No. 5 of OPEN9): $\tau^*(\delta_C) = 27/(4 - \pi^2/49) = 1323/(196 - \pi^2) = \tau^*_{\text{T1c}}$. Two independent symbolic routes lead to one number; this identity is singled out in monograph No. 8 as an independent stability fact.

Corollary 1.4 (the machine reason of uniqueness in v17). Since the pair on the chain reduces to one clock equation (clause (i)), the uniqueness of the non-degenerate static root, established by the Krawczyk census (monograph No. 9), receives an algebraic explanation: looking for two independent clock conditions would be an overdetermination. It is this machine explanation that is named in the ledger the v17 uniqueness reason.

7. Honest Caveats

The theorem is proved for the truncated tower of central amplitudes with the derived dd-law (monograph No. 5). A tower enhancement (the O7 programme, question No. 5 of OPEN9) may refine the value of τ^* , although the agreement of the two independent routes and the ladder structure make a substantial shift unlikely. Puiseux branches at the soft pole $45/8$ (monograph No. 8) do not exclude branch jumps when passing between chain representations; this does not affect the present theorem, since the GCD is computed in the concrete T1c source form.

The second limitation is staticity. All statements of the theorem concern the stationary sector: the live echo dynamics remains with the PDE machine (v6–v9 and campaigns v20–v22), and statement (iv) on the degeneracy of the carrier outside x^* shows precisely the absence of a static carrier, not the absence of dynamics. The third limitation is the conventionality of κ : the closure of $\ln(\tau^*)$ by the additive Berry screening is an observation, not a selection principle (the multiple-testing caveat of campaign v18a); the monograph-sign convention of the multiplicative book moves the estimate away from the observed value (-7.43%), and the choice between the conventions remains open.

Finally, all references to “observed” quantities ($\gamma_{\text{lit}} = 0.374$, $\Delta_{\text{sp}} = 7\pi/30$, $\Delta_{\text{lit}} = 3.44$) are comparison anchors of the repository: no value in the theorem was fitted to them. The defect 0.0 in Table 1.2 refers to the machine arithmetic of the Symbolic layer; the numerical layer (float64) reproduces the same quantities with relative deviations of order 10^{-13} – 10^{-15} , as fixed in the cross-checks of the v14–v15 machines.

8. Reproducibility

The complete reproduction protocol consists of four commands at the repository root (Python ≥ 3.9 , NumPy, SciPy, SymPy, matplotlib, pytest; BLAS is pinned to one thread by the machines automatically). Each command writes a JSON into results/ and is checked by the pytest suite; the corresponding tests are listed in Table 1.5.

Table 1.5. Reproduction protocol for Theorem 1

Step	Command	Result
1	python3 clock_closure_t1c.py	results/clock_closure_t1c.json (layers A0/A1/C1/C2/C3)
2	python3 global_static_search.py	results/global_static_search.json ([G1a] factorization)
3	python3 sympy_center_o7.py	results/q5_center_o7.json (identity of two routes)
4	python3 -m pytest tests/ -q	58 passed (the clock-pair tests included)

The running time of steps 1–3 on an ordinary laptop is minutes; step 4 takes about two minutes. The machines are deterministic: single-threaded BLAS, fixed seeds, offline mode where the saved JSONs suffice. The data of this monograph correspond to the repository state at commit 5b256cd (OPEN9 v19–v22) and to the folder-copy `einstein_direct/open9/`, which reruns autonomously.

References

- [1] Repository wild8highlander/choptuik_ac_bc: einstein_direct/clock_closure_t1c.py; einstein_direct/results/clock_closure_t1c.json (campaign v17, LEDGER §28).
- [2] Ibid.: einstein_direct/global_static_search.py; einstein_direct/results/global_static_search.json (campaign v18, theorem [G1a], LEDGER §29).
- [3] Ibid.: einstein_direct/sympy_center_o7.py; einstein_direct/results/q5_center_o7.json (OPEN9 v19, question No. 5).
- [4] Ibid.: einstein_direct/README.md §§1–3; einstein_direct/README_EN.md §§24–29.
- [5] Ibid.: monograph/verification_monograph_bilingual.pdf — the bilingual verification monograph.
- [6] M. W. Choptuik. Universality and scaling in gravitational collapse of a massless scalar field // Phys. Rev. Lett. 70, 9 (1993) — source of the literature anchor $\gamma_{\text{Ch}} = 0.374$.
- [7] M. D. Roberts. Scalar field counterexamples to the cosmic censorship hypothesis // Gen. Relativ. Gravit. 21, 907 (1989) — the exact solution used as a solver regression.